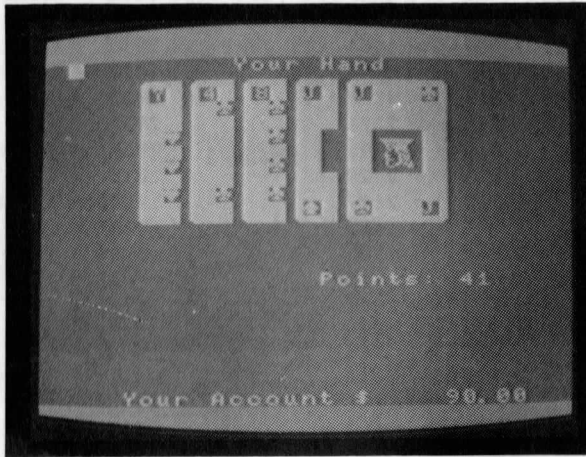


Thirty-One



CLASSIFICATION: Parlor Game

Inflation comes even to Blackjack in this game where the MSX challenges you to a game of cards! Add the value of your cards — if you are closer to 31 than the MSX then you win! Of course, a hand over 31 costs you (or the MSX) the game.

The card values are:

King	13
Queen	12
Jack	11
Ace	1
Others	Face value

PROGRAMMING SUGGESTIONS

The MSX has simple tactics — it stops when it reaches 25. If you have mastered this game, then make the MSX use better tactics, such as remembering what cards have been used and calculating probabilities for the next card. You could even let the MSX cheat a bit by peeking at the next card once in a while (see lines 300-310)!

PROGRAM

Variables

M1\$, M2\$	Messages
CA(I, J)	Card attribute table
DK(I)	Deck of cards
NC	Next card on deck
MB	Money bet this hand
AB	Account balance
PP	Player's points
CP	Computer's points
LC	Location of card
CD	Card to draw at LC
CC	Number of computer's cards
PC	Number of player's cards

Listing

Initialise

```

10 KEY OFF : SCREEN 1, 2 : COLOR 15, 4, 7
15 FOR I = 1 TO 32 : READ Q : A$ = A$ + CHR$( Q ) :
  NEXT : SPRITE$( 0 ) = A$ : A$ = ""
20 FOR I = 1 TO 32 : READ Q : A$ = A$ + CHR$( Q ) :
  NEXT : SPRITE$( 1 ) = A$ : A$ = ""
25 FOR I = 1 TO 32 : READ Q : A$ = A$ + CHR$( Q ) :
  NEXT : SPRITE$( 2 ) = A$ : A$ = ""
30 FOR I = 1072 TO 1143 : READ Q : VPOKE I, Q : NEXT
35 PRINT : PRINT : PRINT "▲▲▲▲▲▲▲▲THIRTY-ONE": VPOKE
  6153, 134 : FOR I = 6154 TO 6165 : VPOKE I, 135 :
  NEXT : VPOKE 6166, 136 : VPOKE 6198, 137 : VPOKE
  6230, 137 : VPOKE 6185, 141 : VPOKE 6217, 141 :
  FOR I = 6282 TO 6293 : VPOKE I, 139 : NEXT
40 VPOKE 6262, 137 : VPOKE 6294, 138 : VPOKE 6249,
  141 : VPOKE 6281, 140
45 FOR I = 384 TO 463 : K = VPEEK( I ) : VPOKE I, INT(
  K / 4 ) : NEXT
50 FOR I = 1 TO 6 : PRINT : NEXT : AB = 100 : PRINT
  "You▲Have▲$100▲to▲St▲▲▲▲art▲With": PRINT : PRINT
  "▲▲▲▲▲▲▲▲Hit▲a▲Key"
55 IF INKEY$ = "" THEN D = RND( 1 ) : GOTO 55
60 CLS : DIM CA ( 52, 2 ) , DK ( 52 )
65 FOR I = 1 TO 4 : FOR J = 1 TO 13 : K = ( I - 1 )
  * 13 + J : CA ( K, 1 ) = J : CA ( K, 2 ) = I : NEXT :
  NEXT : FOR I = 1 TO 52 : DK ( I ) = I : NEXT
75 AB = 100 : NC = 1
80 PRINT TAB( 10 ) "Your▲Hand"
90 PRINT CHR$( 11 ) ; : M1$ = "Please▲wait▲for▲deck▲to▲be
  " : M2$ = "shuffled.▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲": GOSUB 1000 :
  GOSUB 1100 : GOSUB 1200 : IF NC = 1 THEN GOSUB 400
95 GOSUB 300 : GOSUB 200 : GOSUB 200

```

Control

```

100 IF( ( ST ) AND 1 ) = 1 THEN 102 ELSE GOSUB 350
102 IF VPEEK( 6156 ) = 77 THEN 130 ELSE IF( ( ST ) AND
  1 ) <> 1 THEN 110
105 FOR I1 = 1 TO 5 : BEEP : PRINT CHR$( 11 ) TAB( 20 )
  : FOR I2 = 1 TO 300 : NEXT : PRINT CHR$( 11 ) TAB(
  10 ) "MSX▲Hand▲" CHR$( 11 ) ; : NEXT : FOR I = 6176
  TO 6623 : VPOKE I, 32 : NEXT : FOR I = 0 TO 2 :
  PUT SPRITE I, ( 100, 200 ) : NEXT
107 FOR I1 = 1 TO CC : IF I1 < 7 THEN LC = 6179 + 3

```

```

* I1 ELSE LC = 6278 + 3 * ( I1 - 7 )
108 CD = CH ( I1 ) : GOSUB 250 : NEXT : FOR I = 1 TO
15 : PRINT : NEXT : PRINT TAB( 15 ) "Points:"; CP ;
CHR$( I1 ) ;
110 IF( ( ST ) AND 1 ) = 1 THEN 130
120 M1$ = "Your next card-AAAAAAAAAAAA": M2$ = "AAAAAAAAAAAA
AAAAAAAAAAAA": GOSUB 1000
125 GOSUB 200 : IF PP > 31 THEN 500
130 IF( ( ST ) AND 2 ) = 2 THEN 135 ELSE M1$ = "MSX: next c
ardAAAAAAAAAAAA": M2$ = "AAAAAAAAAAAAAAAAAAAAAAAAAAAA":
GOSUB 1000 : FOR T = 1 TO 1500 : NEXT : GOSUB 300 :
IF( CP > 24 AND CP < 32 ) THEN ST = ( ( ST ) OR
2 ) ELSE IF CP > 31 THEN 600
135 IF ST = 3 THEN 450
140 GOTO 100

```

Deal Player

```

200 CD = DK ( NC ) : NC = NC + 1
205 PP = PP + CA ( CD, 1 ) : GOSUB 1100 : PC = PC + 1
210 IF PC < 7 THEN LC = 6179 + 3 * PC ELSE LC = 6278
+ 3 * ( PC - 7 )
215 GOSUB 250
240 RETURN

```

Draw Card On Screen

```

250 VPOKE LC, 134 : FOR I = LC + 1 TO LC + 5 : VPOKE
I, 135 : NEXT : VPOKE LC + 6, 136 : FOR I = LC +
33 TO LC + 289 STEP 32 : FOR J = 0 TO 4 : VPOKE
I - 1, 141 : VPOKE I + 5, 137 : VPOKE I + J, 142 :
NEXT : NEXT
255 VPOKE LC + 320, 140 : FOR I = LC + 321 TO LC + 325 :
VPOKE I, 139 : NEXT : VPOKE LC + 326, 138
260 K1 = CA ( CD, 2 ) : IF K1 = 1 THEN K1 = 131 ELSE
IF K1 = 2 THEN K1 = 129 ELSE IF K1 = 3 THEN K1 =
128 ELSE K1 = 130
265 K2 = CA ( CD, 1 ) : ON K2 GOTO 270, 272, 274, 276,
278, 280, 282, 284, 286, 288, 290, 292, 294
270 VPOKE LC + 33, 65 : VPOKE LC + 163, K1 : VPOKE LC
+ 293, 65 : RETURN
272 VPOKE LC + 33, 50 : VPOKE LC + 67, K1 : VPOKE LC
+ 259, K1 : VPOKE LC + 293, 50 : RETURN
274 VPOKE LC + 33, 51 : VPOKE LC + 67, K1 : VPOKE LC
+ 163, K1 : VPOKE LC + 259, K1 : VPOKE LC + 293,
51 : RETURN
276 VPOKE LC + 33, 52 : VPOKE LC + 293, 52 : VPOKE LC
+ 66, K1 : VPOKE LC + 68, K1 : VPOKE LC + 258, K1 :

```

```

VPOKE LC + 260, K1 : RETURN
278 VPOKE LC + 33, 53 : VPOKE LC + 293, 53 : VPOKE LC
+ 66, K1 : VPOKE LC + 68, K1 : VPOKE LC + 258, K1 :
VPOKE LC + 260, K1 : VPOKE LC + 163, K1 : RETURN
280 VPOKE LC + 33, 54 : VPOKE LC + 293, 54 : VPOKE LC
+ 67, K1 : VPOKE LC + 130, K1 : VPOKE LC + 132,
K1 : VPOKE LC + 194, K1 : VPOKE LC + 196, K1 : VPOKE
LC + 259, K1 : RETURN
282 VPOKE LC + 33, 55 : VPOKE LC + 293, 55 : VPOKE LC
+ 67, K1 : VPOKE LC + 130, K1 : VPOKE LC + 132,
K1 : VPOKE LC + 194, K1 : VPOKE LC + 196, K1 : VPOKE
LC + 258, K1 : VPOKE LC + 260, K1 : RETURN
284 VPOKE LC + 33, 56 : VPOKE LC + 293, 56 : VPOKE LC
+ 66, K1 : VPOKE LC + 68, K1 : VPOKE LC + 130, K1 :
VPOKE LC + 132, K1 : VPOKE LC + 194, K1 : VPOKE
LC + 196, K1 : VPOKE LC + 258, K1 : VPOKE LC + 260,
K1 : RETURN
286 VPOKE LC + 33, 57 : VPOKE LC + 293, 57 : VPOKE LC
+ 163, K1 : VPOKE LC + 66, K1 : VPOKE LC + 68, K1 :
VPOKE LC + 130, K1 : VPOKE LC + 132, K1 : VPOKE
LC + 194, K1 : VPOKE LC + 196, K1 : VPOKE LC + 258,
K1 : VPOKE LC + 260, K1 : RETURN
288 VPOKE LC + 33, 49 : VPOKE LC + 293, 48 : VPOKE LC
+ 292, 49 : VPOKE LC + 34, 48 : VPOKE LC + 99, K1 :
VPOKE LC + 227, K1 : VPOKE LC + 66, K1 : VPOKE LC
+ 68, K1 : VPOKE LC + 130, K1 : VPOKE LC + 132,
K1 : VPOKE LC + 194, K1 : VPOKE LC + 196, K1 : VPOKE
LC + 258, K1 : VPOKE LC + 260, K1 : RETURN
290 I = 8 * INT( ( LC - 6014 ) / 32 ) : J = 8 * ( LC
- 6014 - 32 * I ) : PUT SPRITE 0, ( J + 4, I + 4 )
: FOR I = LC + 130 TO LC + 194 STEP 32 : FOR J =
0 TO 2 : VPOKE I + J, 32 : NEXT : NEXT : VPOKE LC
+ 33, 74 : VPOKE LC + 293, 74 : VPOKE LC + 37, K1 :
VPOKE LC + 289, K1 : RETURN
292 I = 8 * INT( ( LC - 6014 ) / 32 ) : J = 8 * ( LC
- 6014 - 32 * I ) : PUT SPRITE 1, ( J + 4, I + 4 )
: FOR I = LC + 130 TO LC + 194 STEP 32 : FOR J =
0 TO 2 : VPOKE I + J, 32 : NEXT : NEXT : VPOKE LC
+ 33, 81 : VPOKE LC + 293, 81 : VPOKE LC + 37, K1 :
VPOKE LC + 289, K1 : RETURN
294 I = 8 * INT( ( LC - 6014 ) / 32 ) : J = 8 * ( LC
- 6014 - 32 * I ) : PUT SPRITE 2, ( J + 4, I + 4 )
: FOR I = LC + 130 TO LC + 194 STEP 32 : FOR J =
0 TO 2 : VPOKE I + J, 32 : NEXT : NEXT : VPOKE LC
+ 33, 75 : VPOKE LC + 293, 75 : VPOKE LC + 37, K1 :
VPOKE LC + 289, K1 : RETURN
298 RETURN

```

Deal MSX

```

300 CC = CC + 1
305 CD = DK ( NC ) : NC = NC + 1 : CP = CP + CA ( CD,
1 ) : CH ( CC ) = CD
307 IF ( ( ST ) AND 1 ) <> 1 THEN 310 ELSE IF CC < 7
THEN LC = 6179 + 3 * CC ELSE LC = 6278 + 3 * ( CC - 7 )
308 GOSUB 250 : FOR I = 1 TO 15 : PRINT : NEXT : PRINT
TAB ( 15 ) "Points:"; CP ; CHR$ ( 11 ) ;
310 RETURN

```

Bet

```

350 GOSUB 1300 : FOR I = 1 TO 18 : PRINT : NEXT : IF
MB <> 0 THEN 355 ELSE 360
355 INPUT "Change your bet (Y/N)"; X$ : IF X$ = "n" OR
X$ = "N" THEN 365 ELSE INPUT "How much more"; X$ :
IF VAL ( X$ ) > 1000 THEN PRINT CHR$ ( 11 ) ; : GOTO
350 ELSE IF VAL ( X$ ) < 0 THEN PRINT CHR$ ( 11 ) ;
: GOTO 350 ELSE MB = MB + VAL ( X$ ) : AB = AB -
VAL ( X$ ) : GOTO 365
360 INPUT "What is your initial bet"; X$ : IF VAL ( X$ )
> 1000 THEN PRINT CHR$ ( 11 ) ; : GOTO 350 ELSE MB
= VAL ( X$ ) : AB = AB - VAL ( X$ )
365 IF ( ( ST ) AND 1 ) = 0 THEN INPUT "Another card (Y/N)";
X$ : IF X$ <> "y" AND X$ <> "Y" THEN ST = ( ST ) OR 1
370 PRINT CHR$ ( 11 ) ; : GOSUB 1300 : GOSUB 1200 : RETURN

```

Shuffle Deck

```

400 FOR I = 1 TO 100 : J = INT ( RND ( 1 ) * 52 + 1 ) :
K = INT ( RND ( 1 ) * 52 + 1 ) : SWAP DK ( J ) , DK ( K )
410 NEXT
440 RETURN

```

Show Hands

```

450 GOSUB 1300 : FOR I = 1 TO 18 : PRINT : NEXT : PRINT
"MSX has"; CP ; "points": PRINT : K4 = 31 - CP :
K5 = 31 - PP : IF K4 = K5 THEN PRINT "It's a tie!! New
DECK!": AB = AB + MB : GOSUB 400 : CLS : NC = 1 :
GOTO 520
460 FOR T = 1 TO 2000 : NEXT : IF K4 > K5 THEN 600 ELSE
500

```

Player Loses

```

500  PLAY "m59000s816n2r6n3r6n2": GOSUB 1300 : PRINT
    CHR$( 11 ) ; : FOR I = 1 TO 17 : PRINT : NEXT :
    PRINT "YouHave▲lost▲this▲hand": PRINT "Your▲points";
    PP ; ",▲MSX:"; CP : INPUT "Another▲hand▲(Y/N)"; X$
505  IF X$ = "Y" OR X$ = "y" THEN IF NC < 25 THEN PRINT
    "OK▲-▲same▲deck": FOR T = 1 TO 2000 : NEXT : CLS :
    GOTO 520 ELSE PRINT "OK▲-▲new▲deck"; : FOR T = 1
    TO 2000 : NEXT : CLS : NC = 1 : GOTO 520
610  GOTO 900
520  MB = 0 : PP = 0 : ST = 0 : CP = 0 : CC = 0 : PC
    = 0 : FOR I = 0 TO 2 : PUT SPRITE I, ( 100, 200 )
    : NEXT : GOTO 80

```

Player Wins

```

600  PLAY "11s14m3000n44", "11s14m2000n27": GOSUB 1300 :
    PRINT CHR$( 11 ) ; : FOR I = 1 TO 17 : PRINT : NEXT :
    PRINT "You▲Have▲won▲this▲hand": PRINT "Your▲points";
    PP ; ",▲MSX:"; CP : INPUT "Another▲hand▲(Y/N)"; X$
605  IF X$ = "Y" OR X$ = "y" THEN IF NC < 25 THEN PRINT
    "OK▲-▲same▲deck": FOR T = 1 TO 2000 : NEXT : CLS :
    GOTO 620 ELSE PRINT "OK▲-▲new▲deck"; : FOR T = 1
    TO 2000 : NEXT : CLS : NC = 1 : GOTO 620
610  AB = AB + 2 * MB : GOTO 900
620  AB = AB + 2 * MB : MB = 0 : PP = 0 : ST = 0 : CP
    = 0 : CC = 0 : PC = 0 : FOR I = 0 TO 2 : PUT SPRITE
    I, ( 100, 200 ) : NEXT : GOTO 80

```

Game Over

```

900  SCREEN 1 : IF AB < = 0 THEN PRINT "YOU▲DWE▲$"; :
    PRINT USING "#####.##"; - AB : PRINT : PRINT "The▲Colle
    ctors▲will▲be▲": PRINT "visiting▲you▲shortly." : END
910  PRINT "You▲have▲won▲$"; : PRINT USING "#####.##";
    AB : PRINT : PRINT "▲Bet▲you▲can't▲do▲it▲twice!!": END

```

Print Message

```

1000  FOR I = 1 TO 18 : PRINT : NEXT : PRINT M1$ : PRINT :
    PRINT M2$ ; CHR$( 11 ) ;
1010  RETURN

```


Update Points

```
1100 FOR I = 1 TO 15 : PRINT : NEXT : PRINT TAB( 15 )
    "Points:"; PP ; CHR$( 11 ) ;
1110 RETURN
```

Print Account

```
1200 FOR I = 1 TO 23 : PRINT : NEXT : PRINT TAB( 4 )
    "Your Account$"; : PRINT USING "#####.##"; AB ;
    : PRINT CHR$( 11 ) ;
1210 RETURN
```

Blankout Message Area

```
1300 FOR I = 6722 TO 6879 : VPoke I, 32 : NEXT
1310 RETURN
```

Sprite Data

```
10000 DATA 255, 127, 42, 31, 19, 29, 9, 16, 32, 56, 16,
    28, 16, 16, 17, 14, 255, 254, 172, 248, 248, 248,
    188, 136, 116, 92, 84, 108, 106, 218, 151, 7
10010 DATA 0, 5, 11, 15, 31, 62, 60, 121, 120, 120, 120,
    124, 124, 124, 62, 31, 128, 80, 232, 248, 28, 12,
    14, 39, 7, 7, 199, 15, 207, 15, 30, 252
10020 DATA 4, 10, 15, 5, 7, 5, 10, 10, 10, 10, 21, 21,
    26, 54, 73, 48, 68, 170, 254, 84, 252, 72, 136,
    152, 132, 130, 28, 8, 24, 8, 8, 240
```

Character Data

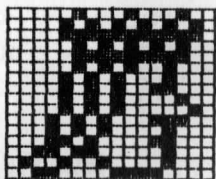
```
10100 DATA 0, 0, 0, 0, 3, 7, 15, 15
10102 DATA 0, 0, 0, 0, 255, 255, 255, 255
10104 DATA 0, 0, 0, 0, 192, 224, 240, 240
10106 DATA 240, 240, 240, 240, 240, 240, 240, 240
10108 DATA 240, 240, 224, 192, 0, 0, 0, 0
10110 DATA 255, 255, 255, 255, 0, 0, 0, 0
10112 DATA 15, 15, 7, 3, 0, 0, 0, 0
10114 DATA 15, 15, 15, 15, 15, 15, 15, 15
10116 DATA 255, 255, 255, 255, 255, 255, 255, 255
```


ChexSum Tables

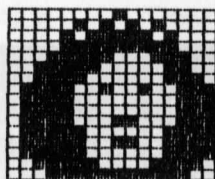
10	= 1215	250	= 11749	460	= 2802
15	= 3995	255	= 4868	500	= 14517
20	= 3996	260	= 6598	505	= 12322
25	= 3997	265	= 4132	510	= 523
30	= 1931	270	= 3099	520	= 6059
35	= 13381	272	= 4014	600	= 15852
40	= 3258	274	= 5330	605	= 12390
45	= 3631	276	= 6514	610	= 1797
50	= 6725	278	= 7843	620	= 7659
55	= 2290	280	= 9481	900	= 10318
60	= 1118	282	= 10867	910	= 6659
65	= 7604	284	= 12206	1000	= 2882
75	= 967	286	= 13803	1010	= 143
80	= 1428	288	= 18778	1100	= 3524
90	= 10893	290	= 17052	1110	= 143
95	= 1071	292	= 17067	1200	= 5546
100	= 1925	294	= 17053	1210	= 143
102	= 3452	298	= 143	1300	= 1682
105	= 12244	300	= 785	1310	= 143
107	= 5026	305	= 4120	10000	= 9134
108	= 5255	307	= 6144	10010	= 8616
110	= 1417	308	= 4084	10020	= 7679
120	= 5276	310	= 143	10100	= 1033
125	= 1534	350	= 3178	10102	= 1384
130	= 16298	355	= 21896	10104	= 1375
135	= 1017	360	= 9849	10106	= 1864
140	= 489	365	= 7154	10108	= 1403
200	= 1636	370	= 1640	10110	= 1416
205	= 2593	400	= 6153	10112	= 1025
210	= 3992	410	= 131	10114	= 1404
215	= 410	440	= 143	10116	= 1888
240	= 143	450	= 14191		

Total= 524162

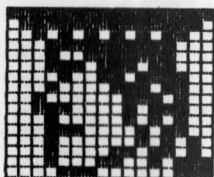
Sprite Shapes



KING

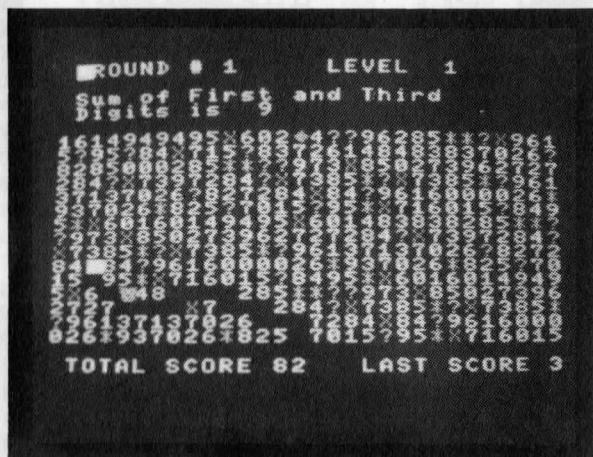


QUEEN



JACK

Minotaur Mastermind



CLASSIFICATION: Educational Game

This is an original strategic game which will test the brain power of even the most experienced masterminds. The challenge is to guess a secret number between 0 and 1000 by working your way through a maze of numbers and symbols, without being gobbled by the minotaur.

Your score is the sum of the digits that you step on — so the **lower** scores are better. When you reach the exit at the top of the maze, you get a chance to guess the secret number. The non-numeric symbols have different effects when stepped on.

- X Supplies a clue
- ? Has random value
- * Randomly alters your score

PROGRAMMING SUGGESTIONS

Minotaur Mastermind could be made into a very complex game by adding more clues (lines 650-708) and different symbols.

You are represented by a flashing point; the controls are:

<P>	-	Up	<:>	-	Down/right
<Ø>	-	Up/left	<@>	-	Right
<;>	-	Down	<L>	-	Down/left
<O>	-	Left	<T>	-	Trapped
<->	-	Up/right			

PROGRAM

Variables

LD	Level of difficulty
R, LS, TS	Round number; last score; total score
SN	Secret number
D1, D2, D3	Digits of secret number
PP	Player's position
M	Minotaur's position
C, Q	For flashing effect
TM	Time
RN, RS	Random number; random number seed
CU, CP	Character under minotaur; character under player
PR, PC	Player's row; player's column
NC	Number of clues
T(I)	Clue I used?
G1, G2, G3	Digits of player's guess
MC, MR	Minotaur's column; minotaur's row

Listing

Initialise

```

5  TM = 1 : INTERVAL ON : ON  INTERVAL = 10 GOSUB 7
6  GOTO 10
7  TM = TM * 2 : RETURN
10 KEY OFF : SCREEN 1 : CLS : COLOR 15, 1, 1 : PRINT
   "▲▲▲▲MINOTAUR▲MASTER-MIND"
15 PRINT : PRINT : PRINT : PRINT "▲LEVEL▲OF▲DIFFICULTY?(1-
   9)";
20 LD = VAL( INKEY$ ) : IF LD < 1 OR LD > 9 THEN 20
   ELSE PRINT LD
22 FOR I = 1088 TO 1344 STEP 64 : FOR J = 0 TO 7 :
   READ Q : VPOKE I + J, Q : NEXT : NEXT
23 VPOKE 8209, 49 : VPOKE 8210, 97 : VPOKE 8211, 177 :
   VPOKE 8212, 145 : VPOKE 8213, 129
25 FOR I = 1 TO 8 : PRINT : NEXT : PRINT "▲▲▲▲HIT▲ANY▲KEY
   ▲TO▲START"
30 IF INKEY$ = "" THEN 30
35 INTERVAL OFF : IF TM > 1000 THEN TM = TM / 3 :
   GOTO 35
40 TS = 0 : NC = 0 : FOR I = 1 TO 10 : T ( I ) = 0 :
   NEXT : R = 1 : SN = INT( TM ) : RS = SN : D1 = INT.(
   SN / 100 ) : D2 = INT( ( SN - D1 * 100 ) / 10 ) :
   D3 = INT( SN - D1 * 100 - D2 * 10 )

```

New Round

```

100 CLS : PP = 6767 : M = 6575
110 FOR I = 6304 TO 6752 STEP 32 : FOR J = 1 TO 30 :
   GOSUB 900 : K1 = INT( RN * 13 ) : IF K1 < 10 THEN
   VPOKE I + J, K1 + 48 : GOTO 130
115 IF K1 = 10 THEN VPOKE I + J, 144 : GOTO 130
120 IF K1 = 11 THEN VPOKE I + J, 152 : GOTO 130
125 VPOKE I + J, 160
130 NEXT : NEXT
140 VPOKE 6767, 48 : LS = 0 : CU = VPEEK( M ) : VPOKE
   M, 136 : VPOKE 6319, 168 : CP = 48
150 PRINT "▲ROUND▲#"; R ; "▲▲▲▲LEVEL▲"; LD ; CHR$( 11 );
160 GOSUB 1000

```

Editor

```

200 C = 0 : Q = 1
205 X$ = INKEY$ : C = C + 1

```

```

210 IF C = 7 THEN Q = 1 - Q : C = 0
215 IF Q <> 0 THEN 230
220 IF VPEEK( PP ) = 255 THEN VPOKE PP, CP : Q = 1 :
    GOTO 230
225 VPOKE PP, 255 : Q = 1
230 IF X$ = " " THEN 205
240 GOSUB 950
242 IF VPEEK( PP ) = 255 THEN VPOKE PP, CP
245 IF X$ = "p" OR X$ = "P" THEN 300
250 IF X$ = "o" OR X$ = "O" THEN 310
255 IF X$ = "@" OR X$ = "." THEN 320
260 IF X$ = ";" OR X$ = "+" THEN 330
265 IF X$ = "-" OR X$ = "=" THEN 340
270 IF X$ = "0" THEN 350
275 IF X$ = "1" OR X$ = "L" THEN 360
280 IF X$ = ":" OR X$ = "*" THEN 370
285 IF X$ = "t" OR X$ = "T" THEN 380
290 GOTO 205

```

Recognise Allowed Moves

```

300 IF VPEEK( PP - 32 ) = 32 THEN 205
302 LS = VPEEK( PP ) - 48
304 VPOKE PP, 32 : PP = PP - 32
305 GOTO 400
310 IF VPEEK( PP - 1 ) = 32 THEN 205
312 LS = VPEEK( PP ) - 48
314 VPOKE PP, 32 : PP = PP - 1
315 GOTO 400
320 IF VPEEK( PP + 1 ) = 32 THEN 205
322 LS = VPEEK( PP ) - 48
324 VPOKE PP, 32 : PP = PP + 1
325 GOTO 400
330 IF VPEEK( PP + 32 ) = 32 THEN 205
332 LS = VPEEK( PP ) - 48
334 VPOKE PP, 32 : PP = PP + 32
335 GOTO 400
340 IF VPEEK( PP - 31 ) = 32 THEN 205
342 LS = VPEEK( PP ) - 48
344 VPOKE PP, 32 : PP = PP - 31
345 GOTO 400
350 IF VPEEK( PP - 33 ) = 32 THEN 205
352 LS = VPEEK( PP ) - 48
354 VPOKE PP, 32 : PP = PP - 33
355 GOTO 400
360 IF VPEEK( PP + 31 ) = 32 THEN 205
362 LS = VPEEK( PP ) - 48
364 VPOKE PP, 32 : PP = PP + 31
365 GOTO 400

```

```

370 IF VPEEK( PP + 33 ) = 32 THEN 205
372 LS = VPEEK( PP ) - 48
374 VPOKE PP, 32 : PP = PP + 33
375 GOTO 400
380 PR = INT( PP / 32 ) : PC = PP - PR * 32 : GOSUB
500 : IF PP <> M THEN 380
385 GOTO 800

```

Check Move

```

400 CP = VPEEK( PP ) : PR = INT( PP / 32 ) : PC = PP
- PR * 32
402 IF( LS > 9 ) OR( LS < 0 ) THEN LS = 0
405 K1 = VPEEK( PP ) : IF K1 = 152 THEN GOSUB 600
410 IF K1 = 144 THEN GOSUB 650
415 IF K1 = 160 THEN GOSUB 750
420 IF K1 = 168 THEN 2000
422 K1 = VPEEK( PP ) - 48 : IF K1 < 0 OR K1 > 9 THEN K1 = 0
425 IF LS + K1 >= 10 - LD THEN GOSUB 500
430 TS = TS + K1 : GOSUB 1000
435 IF PP = M THEN 800
440 GOTO 200

```

Move Minotaur

```

500 VPOKE 8198, 31 : MR = INT( M / 32 ) : MC = M - MR * 32
502 IF MC <> PC THEN 508
504 IF MR > PR THEN MR = MR - 1 : GOTO 540
506 MR = MR + 1 : GOTO 540
508 IF MR <> PR THEN 515
510 IF MC > PC THEN MC = MC - 1 : GOTO 540
512 MC = MC + 1 : GOTO 540
515 IF M > PP THEN 522
517 IF MC < PC THEN MC = MC + 1 : GOTO 540
520 MC = MC - 1 : GOTO 540
522 MR = MR - 1
540 K = 32 * MR + MC : VPOKE 8198, 241 : IF VPEEK( K )
= 131 THEN RETURN
542 VPOKE M, CU
544 M = K : CU = VPEEK( M )
546 VPOKE M, 136
550 PLAY "n2"
570 RETURN

```

Player on (?)

```

600 GOSUB 900 : I = INT( RN * 10 ) : VPOKE PP, I + 48
610 CP = I + 48 : RETURN

```



```

650 VPOKE PP, 48
652 IF NC >= 9 THEN BEEP : GOTO 670
654 GOSUB 900 : K2 = INT( RN * 9 + 1 ) : IF T ( K2 )
    = 1 THEN 654
656 T ( K2 ) = 1 : NC = NC + 1 : BEEP
658 ON K2 GOTO 660, 665, 670, 675, 680, 685, 690,
    695, 700
660 PRINT : PRINT : PRINT "Sum of Digits of Secret":
    PRINT "Number Equals";
662 PRINT D1 + D2 + D3 : PRINT CHR$( 11 ) ; : RETURN
665 PRINT : PRINT : PRINT "Secret Number is";
666 IF SN / 2 = INT( SN / 2 ) THEN PRINT "Even": GOTO 668
667 PRINT "Odd"
668 PRINT CHR$( 11 ) ; : RETURN
670 GOSUB 900 : K = INT( RN * 200 ) + SN : J = K - 200 :
    IF J < 0 THEN J = 0
672 IF K > 1000 THEN K = 1000
673 PRINT : PRINT : PRINT "Number is Between"; J : PRINT
    "and"; K
674 PRINT CHR$( 11 ) ; : RETURN
675 PRINT : PRINT : PRINT "Number is";
676 IF SN / 5 = INT( SN / 5 ) THEN PRINT "Divisible":
    GOTO 678
677 PRINT "Not Divisible"
678 PRINT "by 5": PRINT CHR$( 11 ) ; : RETURN
680 L = D1 : IF D2 > L THEN L = D2
681 IF D3 > L THEN L = D3
682 PRINT : PRINT : PRINT "Largest Digit in the": PRINT
    "Secret Number is"; L ; CHR$( 11 ) ; : RETURN
685 PRINT : PRINT : PRINT "First Digit in the Secret":
    PRINT "Number is";
686 IF D1 > 5 THEN PRINT "Larger Than 5": GOTO 689
687 IF D1 < 5 THEN PRINT "Less Than 5": GOTO 689
688 PRINT "Equal to 5"
689 PRINT CHR$( 11 ) ; : RETURN
690 PRINT : PRINT : PRINT "Sum of First and Third":
    PRINT "Digits is";
692 PRINT D1 + D3 ; CHR$( 11 ) ; : RETURN
695 PRINT : PRINT : PRINT "One of the Digits in the":
    PRINT "Secret Number is";
696 GOSUB 900 : K = INT( RN * 3 ) : IF K = 0 AND SN
    > 99 THEN PRINT D1 : GOTO 699
697 IF K = 1 AND SN > 9 THEN PRINT D2 : GOTO 699
698 PRINT D3
699 PRINT CHR$( 11 ) ; : RETURN

```

```

700 PRINT : PRINT : PRINT "Product of the Non-Zero":
    PRINT "Digits is";
701 IF D1 = 0 AND D2 = 0 THEN PRINT D3 : GOTO 708
702 IF D1 = 0 AND D3 = 0 THEN PRINT D2 : GOTO 708
703 IF D1 = 0 THEN PRINT D2 * D3 : GOTO 708
704 IF D2 = 0 AND D3 = 0 THEN PRINT D1 : GOTO 708
705 IF D2 = 0 THEN PRINT D1 * D3 : GOTO 708
706 IF D3 = 0 THEN PRINT D1 * D2 : GOTO 708
707 PRINT D1 * D2 * D3
708 PRINT CHR$( 11 ) ; : RETURN

```

Player on (*)

```

750 VPOKE PP, 48
752 IF TS < 11 THEN RETURN
754 GOSUB 900 : I = INT( RN * 100 ) : IF I > 50 THEN
    TS = TS - 10 : PLAY "L64N45N46N50N48": RETURN
756 IF I < 10 THEN TS = TS + 50 : PLAY "L32N10N8N6N4N10N8N6
    N4": RETURN
760 TS = TS + 10 : PLAY "L64N10N8N6N4": RETURN

```

Game Over

```

800 CLS : IF M = PP THEN 820
805 FOR I = 1 TO 17 : PRINT "▲▲▲▲CONGRATULATIONS!▲":
    NEXT : PRINT
810 PRINT "YOU▲HAVE▲BEATEN▲THE▲MINOTAUR": PRINT "YOUR▲SCORE
    ▲IS▲": TS
815 PRINT "IN▲": R ; "▲ROUNDS": GOTO 840
820 FOR I = 1 TO 17 : PRINT "MINOTAUR▲WINS▲MINOTAUR▲WINS":
    NEXT : PRINT
825 PRINT "The▲secret▲number▲was▲": SN
840 PRINT : PRINT "HIT▲ANY▲KEY▲FOR▲ANOTHER▲GAME"
845 X$ = INKEY$ : IF X$ = "" THEN 845
846 RUN

```

Random Number

```

900 RN = ( 9999 * RN + RS ) MOD 5997 : RN = RN / 5997!
910 IF RN < .2 THEN RS = RN * 10000 + 1
920 RETURN

```

Blank Out Message

```

950 FOR I = 6208 TO 6271 : VPOKE I, 32 : NEXT
970 RETURN

```

Update Score

```

1000 FOR I = 1 TO 21 : PRINT : NEXT : PRINT "TOTAL▲SCORE"
      TS ; TAB( 17 ) ; "LAST▲SCORE"; LS ; CHR$( 11 ) ;
1010 RETURN

```

Guess Secret Number

```

2000 PRINT : PRINT : PRINT "GUESS▲AT▲THE▲SECRET▲NUMBER"
2005 INPUT "AND▲HIT▲(RETURN)"; K$
2010 I = VAL( K$ ) : IF I = SN THEN 800
2015 G1 = INT( I / 100 ) : G2 = INT( ( I - G1 * 100 )
      / 10 ) : G3 = I - G1 * 100 - G2 * 10
2020 CLS : PRINT : IF D1 = G1 THEN PRINT "First▲Digit▲Correc
      t": PRINT
2025 IF D2 = G2 THEN PRINT "Second▲Digit▲Correct": PRINT
2030 IF D3 = G3 THEN PRINT "Third▲Digit▲Correct"
2035 IF D1 <> G1 AND D2 <> G2 AND D3 <> G3 THEN PRINT
      "No▲Digits▲Correct"
2040 FOR T = 1 TO 2000 : NEXT
2045 R = R + 1 : GOTO 100

```

Character Data

```

10000 DATA 165, 126, 219, 126, 102, 126, 82, 60
10002 DATA 136, 136, 80, 32, 80, 136, 136, 0
10004 DATA 112, 136, 8, 16, 32, 0, 32, 0
10006 DATA 32, 168, 112, 32, 112, 168, 32, 0
10008 DATA 16, 56, 124, 254, 124, 56, 16, 0

```

ChexSum Table

5	= 3376	100	= 1401	215	= 1208
6	= 403	110	= 7235	220	= 3347
7	= 1035	115	= 2238	225	= 1095
10	= 3831	120	= 2239	230	= 1036
15	= 2887	125	= 822	240	= 345
20	= 3348	130	= 320	242	= 2143
22	= 3640	140	= 4161	245	= 1850
23	= 2630	150	= 2754	250	= 1823
25	= 3476	160	= 397	255	= 1838
30	= 969	200	= 734	260	= 1779
35	= 3770	205	= 1325	265	= 1771
40	= 12987	210	= 1798	270	= 977

275	= 1886	500	= 2962	696	= 4300
280	= 1802	502	= 1423	697	= 2351
285	= 1914	504	= 2344	698	= 266
290	= 594	506	= 1325	699	= 932
300	= 1780	508	= 1212	700	= 4374
302	= 1398	510	= 2282	701	= 2407
304	= 1349	512	= 1294	702	= 2407
305	= 537	515	= 884	703	= 2127
310	= 1791	517	= 2286	704	= 2408
312	= 1398	520	= 1293	705	= 2125
314	= 1365	522	= 831	706	= 2124
315	= 537	540	= 3410	707	= 1018
320	= 1791	542	= 481	708	= 932
322	= 1398	544	= 1424	750	= 477
324	= 1364	546	= 481	752	= 947
325	= 537	550	= 431	754	= 5284
330	= 1780	570	= 143	756	= 3807
332	= 1398	600	= 2558	760	= 2259
334	= 1348	610	= 981	800	= 1151
335	= 537	650	= 477	805	= 3152
340	= 1814	652	= 1838	810	= 4559
342	= 1398	654	= 3251	815	= 1883
344	= 1380	656	= 1724	820	= 3962
345	= 537	658	= 3150	825	= 2729
350	= 1824	660	= 4854	840	= 2697
352	= 1398	662	= 2064	845	= 1602
354	= 1396	665	= 2451	846	= 139
355	= 537	666	= 3357	900	= 3412
360	= 1814	667	= 497	910	= 2090
362	= 1398	668	= 932	920	= 143
364	= 1379	670	= 4489	950	= 1584
365	= 537	672	= 1542	970	= 143
370	= 1824	673	= 3430	1000	= 5315
372	= 1398	674	= 932	1010	= 143
374	= 1395	675	= 1696	2000	= 2804
375	= 537	676	= 4001	2005	= 1608
380	= 4638	677	= 1548	2010	= 1876
385	= 391	678	= 1556	2015	= 5740
400	= 3919	680	= 1767	2020	= 3830
402	= 2128	681	= 1251	2025	= 3499
405	= 2306	682	= 5960	2030	= 3150
410	= 1197	685	= 4644	2035	= 5189
415	= 1313	686	= 2960	2040	= 1157
420	= 1162	687	= 2739	2045	= 1250
422	= 3290	688	= 1125	10000	= 1770
425	= 2245	689	= 932	10002	= 1572
430	= 1429	690	= 4204	10004	= 1340
435	= 868	692	= 1513	10006	= 1567
440	= 593	695	= 5220	10008	= 1517

Total= 376804

Appendix A

Using Joysticks

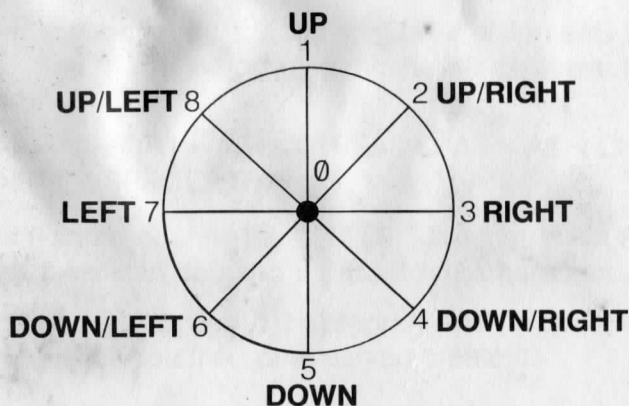
To convert keyboard games to joystick games, you will need to replace USR calls to location 60000 (which reads the keyboard) with a BASIC sub-routine similar to the one below. [Use STICK(1) to read joystick 1 or STICK(2) for joystick 2; STICK(0) reads the keyboard.]

```
A = STICK(1)
IF (A = 0) THEN RETURN [to main program]
IF (A = 1) OR (A = 2) OR (A = 8) THEN [move up]
IF (A = 8) OR (A = 7) OR (A = 6) THEN [move left]
IF (A = 6) OR (A = 5) OR (A = 4) THEN [move down]
IF (A = 2) OR (A = 3) OR (A = 4) THEN [move right]
```

These 'move' statements could be USR calls to location 60118 accompanied by appropriate pokes (see Appendix C, 'Machine Code Support Program').

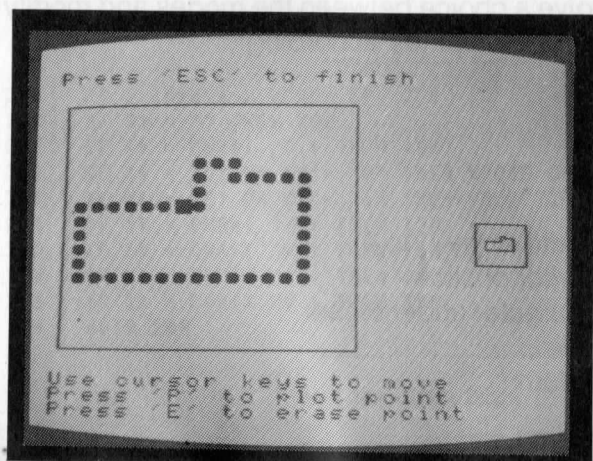
To alter firing, simply change any references to STRIG(0) to STRIG(1) for joystick 1, or STRIG(2) for joystick 2.

JOYSTICK VALUES, STRIG(n)



Appendix B

Sprite Maker



This program makes the creation of sprites a very easy task. You create a sprite on a large grid by moving the cursor around and plotting points. When you are finished, the program supplies you with the numbers that you need to create the sprite in your own program.

If these numbers are placed in DATA statements in your program, then the following statements will create a 16×16 sprite:

```
10 A$ = " ": FOR I = 1 TO 32: READ Q: A$ = A$ + CHR$(Q):  
NEXT I: SPRITE$(O) = A$
```

For an 8×8 sprite, change the '32' to '8'. Note that, in 8×8 mode, the sprite maker can be used for redefining characters.

Instructions are included in the program and the box to the right of the grid shows you the sprite as you create it.

PROGRAMMING SUGGESTIONS

The colour of the sprite can be set by adding the colour number to the end of the 'PUT SPRITE' statement in lines 120 and 520.

If you want to see your sprite in the enlarged mode, you could make the program give a choice between the modes and modify the boxes surrounding the sprites accordingly.

PROGRAM Variables

SA(I, J)	Sprite Array (1 to 0)
CP	Cursor position
CU	Character under cursor
RW	Row
CO	Column

LISTING

Initialise

```
10  KEY OFF : COLOR 1, 15, 15 : SCREEN 1, 2 : PRINT
    "■■■■■■■■SPRITE▲MAKER"
15  PRINT : PRINT : PRINT : INPUT "Choose▲8x8▲('8')▲or■■■■■■
    ▲■■■■■■■■16x16▲('16')"; X$
20  IF X$ = "16" THEN 500 ELSE IF X$ = "8" THEN 100 ELSE
    PRINT "Pardon?": GOTO 15
```

Set-up Screen (8 × 8 sprite)

```
100  CLS : PRINT "▲▲▲Press▲'ESC'▲to▲finish": FOR I =
    1 TO 5 : PRINT : NEXT
105  PRINT "▲▲+-----+": FOR I = 1 TO 8 : PRINT "▲▲!▲▲▲
    ▲▲!": NEXT : PRINT "▲▲+-----+"
110  PRINT : PRINT : PRINT "▲▲▲Use▲cursor▲keys▲to▲move":
    PRINT : PRINT "▲▲▲Press▲'P'▲to▲plot▲point": PRINT :
```



```

PRINT "▲▲▲Press▲'E'▲to▲erase▲point"
115 CP = 6473 : CU = 32 : RW = 4 : CD = 4
120 A$ = "": SPRITE$( 0 ) = A$ : PUT SPRITE 0, ( 167, 86 )
125 VPOKE 6484, 24 : VPOKE 6485, 23 : VPOKE 6486, 25 :
VPOKE 6516, 22 : VPOKE 6518, 22 : VPOKE 6549, 23 :
VPOKE 6548, 26 : VPOKE 6550, 27

```

Control (8 × 8)

```

200 X$ = INKEY$ : VPOKE CP, 255
250 IF X$ = "" THEN 200
255 IF X$ = CHR$( 27 ) THEN 400
260 IF X$ = "p" OR X$ = "P" THEN VPOKE CP, CU : GOTO 300
265 IF X$ = "E" OR X$ = "e" THEN VPOKE CP, CU : GOTO 310
270 IF X$ = CHR$( 30 ) THEN 320
275 IF X$ = CHR$( 29 ) THEN 330
280 IF X$ = CHR$( 31 ) THEN 340
285 IF X$ = CHR$( 28 ) THEN 350
290 GOTO 200

```

Move/Plot/Erase

```

300 IF VPEEK( CP ) = 133 THEN 200 ELSE CU = 133 : SA
( RW, CO ) = 1 : GOTO 380
310 IF VPEEK( CP ) = 32 THEN 200 ELSE CU = 32 : SA (
RW, CO ) = 0 : GOTO 380
320 IF RW < 2 THEN 200 ELSE RW = RW - 1 : VPOKE CP,
CU : CU = VPEEK( CP - 32 ) : CP = CP - 32 : GOTO 200
330 IF CO < 2 THEN 200 ELSE CO = CO - 1 : VPOKE CP,
CU : CU = VPEEK( CP - 1 ) : CP = CP - 1 : GOTO 200
340 IF RW > 7 THEN 200 ELSE RW = RW + 1 : VPOKE CP,
CU : CU = VPEEK( CP + 32 ) : CP = CP + 32 : GOTO 200
350 IF CO > 7 THEN 200 ELSE CO = CO + 1 : VPOKE CP,
CU : CU = VPEEK( CP + 1 ) : CP = CP + 1 : GOTO 200

```

Print Sprite (8 × 8)

```

380 K2 = 14335 + RW : K = VPEEK( K2 ) : K1 = 2 ^ ( 8
- CO ) : IF SA ( RW, CO ) = 1 THEN K = K + K1 :
VPOKE K2, K : GOTO 200
385 K = K - K1 : VPOKE K2, K : GOTO 200

```


Finish (8 × 8)

```

400 CLS : PRINT "Numbers▲for▲'DATA'▲statements": PRINT :
    PRINT : PRINT
410 FOR I = 14336 TO 14342 : PRINT USING "###"; VPEEK( I )
    ; : PRINT "▲"; : NEXT
415 PRINT : PRINT : PRINT : PRINT USING "###"; VPEEK(
    14343 )
420 FOR I = 1 TO 4 : PRINT : NEXT : PRINT "Hit▲any▲key▲to▲
    start▲again" : PUT SPRITE 0, ( 127, 150 )
430 IF INKEY$ = "" THEN 430 ELSE RUN

```

Set-up Screen (16 × 16)

```

500 CLS : PRINT "Press▲'ESC'▲to▲finish": PRINT
505 PRINT "+-----+": FOR I = 1 TO 16 : PRINT
    "#####": NEXT : PRINT "+-----+"
510 PRINT : PRINT "Use▲cursor▲keys▲to▲move": PRINT "Press▲
    P'▲to▲plot▲point": PRINT "Press▲'E'▲to▲erase▲point";
515 DIM SA ( 16, 16 ) : CP = 6474 : CU = 32 : RW = 8 :
    CO = 8
520 A$ = "": SPRITE$( 0 ) = A$ : PUT SPRITE 0, ( 216, 86 )
525 VPOKE 6490, 24 : VPOKE 6491, 23 : VPOKE 6492, 23 :
    VPOKE 6493, 25 : VPOKE 6522, 22 : VPOKE 6525, 22 :
    VPOKE 6554, 22 : VPOKE 6557, 22 : VPOKE 6587, 23 :
    VPOKE 6588, 23 : VPOKE 6586, 26 : VPOKE 6589, 27

```

Control (16 × 16)

```

600 X$ = INKEY$ : VPOKE CP, 255
650 IF X$ = "" THEN 600
655 IF X$ = CHR$( 27 ) THEN 800
660 IF X$ = "p" OR X$ = "P" THEN VPOKE CP, CU : GOTO 700
665 IF X$ = "e" OR X$ = "E" THEN VPOKE CP, CU : GOTO 710
670 IF X$ = CHR$( 30 ) THEN 720
675 IF X$ = CHR$( 29 ) THEN 730
680 IF X$ = CHR$( 31 ) THEN 740
685 IF X$ = CHR$( 28 ) THEN 750
690 GOTO 600

```

Move/Plot Erase (16 × 16)

```

700 IF VPEEK( CP ) = 133 THEN 600 ELSE CU = 133 : SA
    ( RW, CO ) = 1 : GOTO 780

```

```

710 IF VPEEK( CP ) = 32 THEN 600 ELSE CU = 32 : SA (
RW, CO ) = 0 : GOTO 780
720 IF RW < 2 THEN 600 ELSE RW = RW - 1 : VPOKE CP,
CU : CU = VPEEK( CP - 32 ) : CP = CP - 32 : GOTO 600
730 IF CO < 2 THEN 600 ELSE CO = CO - 1 : VPOKE CP,
CU : CU = VPEEK( CP - 1 ) : CP = CP - 1 : GOTO 600
740 IF RW > 15 THEN 600 ELSE RW = RW + 1 : VPOKE CP,
CU : CU = VPEEK( CP + 32 ) : CP = CP + 32 : GOTO 600
750 IF CO > 15 THEN 600 ELSE CO = CO + 1 : VPOKE CP,
CU : CU = VPEEK( CP + 1 ) : CP = CP + 1 : GOTO 600

```

Print Sprite (16 × 16)

```

780 IF CO < 9 THEN K1 = 0 ELSE K1 = 16
785 K2 = 14335 + RW + K1 : K = VPEEK( K2 ) : IF CO <
9 THEN K1 = 2 ^ ( 8 - CO ) ELSE K1 = 2 ^ ( 16 - CO )
790 IF SA ( RW, CO ) = 1 THEN K = K + K1 : VPOKE K2,
K : GOTO 600
795 K = K - K1 : VPOKE K2, K : GOTO 600

```

Finish (16 × 16)

```

800 CLS : PRINT "Numbers▲for▲'DATA'▲statements": PRINT
"▲Read▲from▲left▲to▲right": PRINT
810 FOR I = 14335 TO 14356 STEP 7 : PRINT : PRINT :
FOR J = 1 TO 7 : PRINT USING "###"; VPEEK( I + J )
; : PRINT "▲"; : NEXT : NEXT
815 PRINT : PRINT : FOR I = 14364 TO 14367 : PRINT USING
"###"; VPEEK( I ) ; : PRINT "▲"; : NEXT
820 FOR I = 1 TO 3 : PRINT : NEXT : PRINT "▲Hit▲any▲key▲to▲
start▲again" : PUT SPRITE 0, ( 120, 165 )
830 IF INKEY$ = "" THEN 830 ELSE RUN

```

ChexSum Table

10	= 3219	320	= 6213	665	= 3076
15	= 4148	330	= 6174	670	= 1516
20	= 4379	340	= 6210	675	= 1516
100	= 4028	350	= 6171	680	= 1532
105	= 4018	380	= 7500	685	= 1532
110	= 10384	385	= 2007	690	= 483
115	= 2034	400	= 4116	700	= 4051
120	= 2258	410	= 2988	710	= 3750
125	= 5134	415	= 1934	720	= 5975
200	= 1337	420	= 5353	730	= 5936
250	= 1032	430	= 1484	740	= 6031
255	= 1449	500	= 2698	750	= 6008
260	= 2947	505	= 6460	780	= 1845
265	= 2916	510	= 8747	785	= 6451
270	= 1373	515	= 2718	790	= 3234
275	= 1373	520	= 2297	795	= 1879
280	= 1389	525	= 9318	800	= 6481
285	= 1389	600	= 1337	810	= 5442
290	= 593	650	= 920	815	= 3392
300	= 4243	655	= 1292	820	= 5345
310	= 3938	660	= 3075	830	= 1374

Total= 229442

Appendix C

Machine-Code Support Program

SUPPORT PROGRAM

This program is required for a number of the games in this book. It must be loaded and RUN before you load your game program.

- 1 Type this program in and save it. Use ChexSum to check it and, when bug-free, save it as "SUPPRT".
- 2 Load Support and RUN it.
- 3 Load your debugged game program and RUN it; it will call and use support as required.

Support Listing

```
10  CLS : PRINT "MACHINE CODE SUPPORT PROGRAM"
20  FOR I = 60000! TO 60217! : READ Q : POKE I, Q : NEXT
1000 DATA 219, 170, 230, 240, 33, 92, 234, 94, 246, 8,
      0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 211, 170, 219,
      169, 71, 0, 0, 0, 0, 0, 0, 0, 0, 0
1010 DATA 0, 0, 0, 230, 128, 194, 151, 234, 203, 91,
      202, 151, 234, 62, 3, 50, 94, 234, 195, 209, 234,
      120, 230, 64, 194, 170, 234, 203, 83, 202, 170,
      234, 62, 2
1020 DATA 50, 94, 234, 195, 209, 234, 120, 230, 32, 194,
      189, 234, 203, 67, 202, 189, 234, 62, 0, 50, 94,
      234, 195, 209, 234, 120, 230, 16, 194, 208, 234,
      203, 75, 202, 208, 234, 62, 1, 50, 94, 234, 195,
      209, 234, 201, 62, 0, 50, 93, 234, 58
1030 DATA 93, 234, 203, 39, 203, 39, 79, 6, 0, 33, 0,
      27, 9, 58, 95, 234, 71, 58, 94, 234, 87, 254, 0,
      194, 250, 234, 205, 27, 235, 121, 152, 79, 195,
      44, 235, 254
1040 DATA 1, 194, 6, 235, 35, 205, 27, 235, 195, 244,
      234, 254, 2, 194, 20, 235, 205, 27, 235, 121, 128,
      79, 195, 44, 235, 35, 205, 27, 235, 195, 14, 235,
      125, 243, 211
```

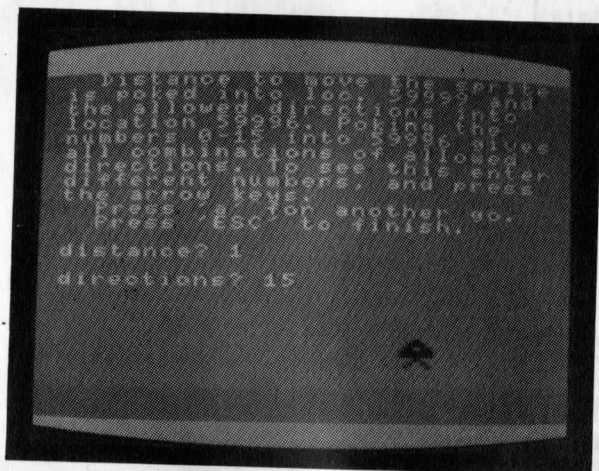
1050 DATA 153, 124, 211, 153, 0, 0, 0, 0, 219, 152, 79,
183, 251, 201, 125, 243, 211, 153, 124, 246, 64,
211, 153, 121, 211, 152, 251, 201

Support ChexSum Table

10 = 2712
20 = 2233
1000 = 7068
1010 = 10317
1020 = 21367
1030 = 10298
1040 = 11087
1050 = 8282

Total= 73364

DEMONSTRATION PROGRAM



This program gives an explanation and demonstration of the machine-code support program used to move sprites in many of the games in this book. It is divided into two parts, Part 1 involving the reading of the keyboard, and Part 2 the moving of sprites.

All instructions are included in the program.

Demonstration Program Suggestions

Part 2 only allows the user to choose directions for moving the sprite. The '1' in line 90 can be altered to give different speeds, or you can restructure the program to give the user the choice of speeds when the program runs.

Demonstration Variables

X\$	Key Pressed
K	Temporary

Demonstration Listing

Initialise

```
5  REM RUN MACHINE CODE
6  REM SUPPORT PROGRAM
7  REM SEE APPENDICES
10 COLOR 15, 4, 7 : SCREEN 1, 1 : KEY OFF
15 FOR I = 1 TO 8 : READ Q : A$ = A$ + CHR$( Q ) :
NEXT : SPRITE$( 0 ) = A$ : GOTO 200
```

Part 1: Read Keyboards

```
20 CLS : PRINT "▲▲DEMONSTRATION▲OF▲MACHINE": PRINT
  "▲▲▲▲CODE▲SUPPORT▲PROGRAM": PRINT : PRINT "▲▲▲▲▲▲▲▲▲▲▲▲
ART▲1": PRINT
25 DEFUSR0 = 60000! : PRINT "▲▲The▲first▲routine▲starts▲at
location▲60000.▲Its▲purpose": PRINT "is▲to▲read▲the▲key
board▲and▲": PRINT "move▲sprite▲0▲if▲the▲arrow":
PRINT "keys▲are▲pressed."
30 PRINT "▲▲The▲4▲allowed▲directions": PRINT "are▲numbered
as▲follows": PRINT
35 PRINT "▲▲▲▲▲▲▲▲▲▲▲0": PRINT : PRINT "▲▲▲▲▲▲▲▲▲▲▲U":
PRINT "▲▲▲▲▲▲▲▲▲▲▲V": PRINT "▲▲▲▲▲▲▲▲1▲UWU▲3":
PRINT "▲▲▲▲▲▲▲▲▲▲▲V": PRINT "▲▲▲▲▲▲▲▲▲▲▲U":
PRINT : PRINT "▲▲▲▲▲▲▲▲▲▲▲2": PRINT : PRINT "▲▲Hit▲an
y▲key▲to▲continue";
40 IF INKEY$ = "" THEN 40
45 CLS : PRINT "▲▲Distance▲to▲move▲the▲spriteis▲poked▲into
▲loc.▲59999▲and": PRINT "the▲allowed▲directions▲into":
PRINT "location▲59996.▲Poking▲the▲": PRINT "numbers▲0-1
5▲into▲59996▲gives";
50 PRINT "all▲combinations▲of▲allowed": PRINT "directions.
▲To▲see▲this▲enterdifferent▲numbers,▲and▲press":
PRINT "the▲arrow▲keys.": PRINT "▲▲Press▲'a'▲for▲another
▲go.": PRINT "▲▲Press▲'ESC'▲to▲finish.": PRINT
55 INPUT "distance": D : PRINT : INPUT "directions": X$
60 PRINT : PRINT : PUT SPRITE 0, ( 120, 175 ), 1
65 POKE 59999!, D : POKE 59996!, VAL( X$ ) : I = USR0(
I ) : A$ = INKEY$ : IF A$ = "a" OR A$ = "A" THEN
55 ELSE IF A$ = CHR$( 27 ) THEN PUT SPRITE 0, (
100, 200 ) : GOTO 200 ELSE 65
```


Part 2: Move Sprites

```

75  CLS : PRINT "#####PART 2": PRINT : PRINT "##This routine starts at loc.": PRINT "60118 and moves a sprite.": PRINT "The sprite number must be a poked into 59997, direction into 59998 and distance into 59999."
80  PRINT "##Hit 'RETURN' to change a direction.": PRINT "##Hit 'ESC' to finish.": PRINT
85  DEFUSR1 = 60118! : POKE 59997!, 0 : PUT SPRITE 0, ( 10, 175 ), 1
90  POKE 59998!, 1 : POKE 59999!, 1
95  I = USR1( I ) : X$ = INKEY$ : IF X$ = "" THEN 95
    ELSE IF X$ = CHR$( 27 ) THEN PUT SPRITE 0, ( 100, 200 ) : GOTO 200 ELSE IF X$ <> CHR$( 13 ) THEN 95
96  INPUT "Which direction (0-3)"; X$ : K = VAL( X$ ) : IF K < 0 OR K > 3 THEN PRINT "Pardon?": GOTO 96
    ELSE POKE 59998!, K : CLS : PRINT "##Hit 'RETURN' to change a direction.": PRINT "##Hit 'ESC' to finish.": GOTO 95

```

Menu

```

200  CLS : PRINT "MACHINE CODE DEMONSTRATION"
205  FOR I = 1 TO 4 : PRINT : NEXT : PRINT TAB( 18 )
    "Press Key": PRINT : PRINT : PRINT "#### DEMO PART 1 ####"
    "1": PRINT : PRINT : PRINT "#### DEMO PART 2 ####"
    "2": PRINT : PRINT : PRINT "#### FINISH ####"
210  FOR I = 1 TO 4 : PRINT : NEXT : PRINT "Press one of the above keys": PRINT "to make your selection"
220  X$ = INKEY$ : IF X$ = "1" THEN 20 ELSE IF X$ = "2" THEN 75 ELSE IF X$ = "F" OR X$ = "f" THEN SCREEN 1 : PRINT "#### GOOD-BYE!": END ELSE 220
1000 DATA 24, 60, 102, 255, 255, 24, 36, 66

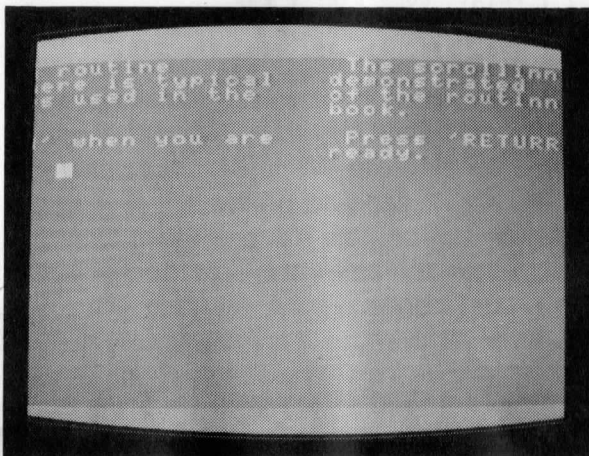
```

Demonstration ChexSum Table

5	= 0	40	= 979	90	= 1523
6	= 0	45	= 20547	95	= 8437
7	= 0	50	= 23681	96	= 19288
10	= 1235	55	= 3192	200	= 2619
15	= 4280	60	= 1400	205	= 13114
20	= 8057	65	= 10795	210	= 7199
25	= 19414	75	= 24879	220	= 8044
30	= 5884	80	= 6587	1000	= 1583
35	= 20681	85	= 2879	Total	= 216297

Appendix D

Machine-Code Scrolling Routine



There are several machine-code scrolling routines used in this book, but all are very similar to the one used in this demonstration. All instructions are included in the program.

PROGRAM

Listing of BASIC Loader

```
5 REM RUN MACHINE CODE
6 REM SUPPORT PROGRAM
7 REM SEE APPENDICES
10 SCREEN 1 : COLOR 15, 4, 7 : KEY OFF : FOR I = 60350!
    TO 60401! : READ Q : POKE I, Q : NEXT
15 DEFUSR = 60350! : POKE 60346!, 0 : POKE 60347!,
    24 : POKE 60348!, 255 : POKE 60349!, 26
```

```

20 PRINT "▲▲▲DEMONSTRATION▲OF▲MACHINE": PRINT "▲▲▲CODE▲SC
   ROLLING▲ROUTINE"
25 FOR I = 1 TO 10 : PRINT : NEXT : PRINT "▲▲▲Press▲any▲k
   ey▲to▲start": PRINT : PRINT "▲▲▲Press▲'ESC'▲to▲finish"
30 IF INKEY$ = "" THEN 30
40 CLS : PRINT "▲▲The▲scrolling▲routine▲▲▲▲▲demonstrated
   ▲here▲is▲typical▲of▲the▲routines▲used▲in▲the▲book.▲"
45 PRINT : PRINT "▲▲Press▲'RETURN'▲when▲you▲are▲ready."
50 IF INKEY$ = CHR$( 13 ) THEN 55 ELSE 50
55 D = USR( D ) : X$ = INKEY$ : IF X$ = CHR$( 27 )
   THEN SCREEN 1 : PRINT "GOOD-BYE!": END
60 GOTO 55
10000 DATA 6, 31, 42, 186, 235, 43, 35, 205, 27, 235,
        120, 254, 31, 202, 215, 235, 4, 43, 205, 44, 235,
        35, 195, 227, 235, 6, 0, 17, 31, 0, 25, 205, 44,
        235, 183, 237, 82, 237, 91, 188, 235, 123, 189,
        194, 196, 235, 122, 188, 194, 196, 235, 201

```

ChexSum Table

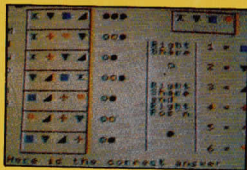
5	= 0
6	= 0
7	= 0
10	= 3647
15	= 4235
20	= 5324
25	= 6972
30	= 969
40	= 10464
45	= 3749
50	= 1782
55	= 4360
60	= 444
10000	= 21635

Total= 63581

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